

EVA-1 from KK System – Reproduction

User Manual

v1.0

1 Connection

1.1 Serial connection:

The connection between HX-20 and EVA-1 must be with a 5-pin DIN “crossover” cable. Pinout of the cable:

Pin HX20	Pin EVA1	Function	Comment
1	1	GND	
2	3	PTX Out (Transmit Data)	
3	2	PRX In (Receive Data)	
4	5	Pout Out (Transmit Mode)	not used
5	4	Pin In (Receive Mode)	not used

The cable must be connected to the left socket. The right socket is for a portable floppydrive.

In the new Version of the Board is a jumper (JP8). When this Jumper is closed, a “straight 5-pin DIN cable can be used. This cable must be connected to the right socket. In this case a portable floppydrive cannot be used!

1.2 Monitor Connection:

There are two monitor connections, a Video connector (FBAS, SCART compatible) and a CGA Type connection (15.6kHz, 50 Hz) for a CGA compatible monitor.

2 Usage

After connection the monitor and power the startup screen should show. Here are the most common shortcuts and also the first entry for the HX-20.

2.1 Control characters:

CTRL+E	Deletes from Cursor to the end of the line
CTRL+F	Toggle Inverse (only for “special” special characters)
CTRL+G	Toggle bright (only for “special” special characters)
CTRL+N	“special” character from now on. This writes every character with Bit 7 set. It won’t show unless one “activates” them with CTRL+F or CTRL+G
CTRL+I	Horizontal TAB
CTRL+P	Move Window up. The TEXTRAM Chip on the Reproduction board is twice as big. It can store 50 Lines with 80 characters. Moves only the Window of the screenbuffer.
CTRL+Q	Move Window down. Moves only the Window of the screenbuffer.
CTRL+T	Restores the original screenbuffer window.
CTRL+S	Jumps into monitor program. In the new EPROM is a monitor program (MC3 monitor from Daniel Tufvesson). After CTRL+S connect a terminal (38400Baud) and press Enter.
CTRL+V	Cursor visible
CTRL+W	Cursor invisible
CTRL+Z	Deletes from Cursor to the end of the screen
CTRL+0	Show testpicture (works only in SCREEN 1,1 Mode)
CTRL+1	toggle Text ON/OFF
CTRL+2	toggle Inverted screen
CTRL+3	switch to 32 characters / Line
CTRL+4	switch to 40 characters / Line
CTRL+5	toggle Cursor BLINK
CTRL+6	Clear graphics screen (GCLS)
CTRL+7	switch to 72 characters / Line
CTRL+8	switch to 80 characters / Line
CTRL+9	toggle character set (the second character set has only minor differences – see Reference Manual Appendix F)

3 EPSP Functions

The EVA-1 handles the following EPSP Functions:

\$88	SCRNSIZE	READ SCREEN SIZE (L.C)
\$89	SCRNSIZE	GET PHYSICAL SCREEN SIZE (L.C)
\$8C	GCRSRPOS	GET CURSOR POSITION
\$8F	GETPOINT	GET POINT ON THE DISPLAY (L.C)
\$91	READLINE	READ THE EXTEND OF CURRENT LINE (L.C)
\$92	WRITECHAR	WRITE ONE CHARACTER TO VIRTUAL SCREEN (L.C)
\$93	SETDISPMOD	SET DISPLAY MODE (C)
\$97	READCHAR	READ CHARACTERS FROM VS. (L,C)
\$98	WRITECHAR	WRITE ONE CHARACTER TO VS AND GET EXTEND OF NEW CURSOR POSITION.(L,C)
\$C2	SETCRSR	SET CURSOR POSITION ON THE VIRTUAL SCREEN (L.C)
\$C5	SRLISTFLG	SET LIST FLAG (L.C)
\$C6	SRLISTFLG	RESET LIST FLAG (L.C)
\$C7	SETPOINT	SET POINT TO DISPLAY (L.C)
\$C8	DRAWLINE	DRAW LINE TO DISPLAY (L.C)
\$C9	SETLINETERM	SET LINE TERMINATE POSITION (L.C)
\$CA	CLRGRAPH	CLEAR THE GRAPHICS SCREEN. (L.C)
\$CB	DISABLED	SET SCRLUPSPEED (C)
\$CF	DISABLED	COLOR SET SELECT (C)
\$D4	DISABLED	?

